

Spatial Columns Fact Sheet

A one-page reference for citing when SQL Server spatial geometry was added to the Viking annotation database. Prepared for the 3D connectome analysis methods paper.

When were spatial columns added?

The `MosaicShape` and `VolumeShape` spatial geometry columns were added to the `Location` table on **19 August 2015**, in commit `ffe9cceb` on the dev branch (“Migration to spatial columns in database”, author James Anderson). That is roughly **11 years** before the present (June 2026) and about **6 years after** the 2009 PLoS Biology framework paper (Anderson et al.).

The change is recorded in the versioned database build script, `Servers/SQL/DatabaseCreateUpdate/CreateUpdateDatabase.sql` (originally `Servers/AnnotationService/SQL/DatabaseCreateUpdate/CreateUpdateDatabase.sql` before later repository reorganization).

What changed (two-step migration)

The migration is applied as two sequential schema versions in the `DBVersion` table:

DBVersion	Date	Change
24	2015-08-19	ALTER TABLE Location ADD MosaicShape geometry and ADD VolumeShape geometry Populate the new shapes from the legacy scalar columns; drop
25	2015-08-19	Vertices, X, Y, VolumeX, VolumeY; re-add X/Y/VolumeX/VolumeY as persisted computed centroids of the shapes

Before this migration, an annotation footprint was stored as scalar columns (`X`, `Y`, `Radius`) plus a `Vertices` vertex list for polygons. After the migration, each `Location` row stores two true geometries:

- `MosaicShape` – the footprint in the native section/mosaic coordinate space.
- `VolumeShape` – the same footprint in the registered volume coordinate space.

Circular annotations were converted to `CURVEPOLYGON(CIRCULARSTRING(...))` and point annotations to `POINT(...)`, as seen in the version 25 block of the build script (around lines 2389-2440). The scalar centroid columns are retained as `PERSISTED` computed columns, for example:

```
ALTER TABLE Location ADD X as ISNULL(MosaicShape.STCentroid().STX, ISNULL(
```

Follow-up spatial commits (same era)

Commit	Date	Summary
ffe9cceb	2015-08-19	Migration to spatial columns in database (DBVersion 24-25)
b579196a	2015-08-21	Further updates for spatial migration
19b70b34	2015-08-26	Add spatial index to the database
64203551	2016-03-24	Spatial queries use mosaic or volume coordinates; synapse-to-lines migration

The same script has since grown to **DBVersion 83**, which improves the spatial bounds stored procedure `SelectSectionLocationsAndLinksInMosaicBounds`.

Methods-ready paragraph (copy/paste)

Annotation geometry in the Viking database was migrated from scalar coordinates (X, Y, radius, and polygon vertex lists) to native SQL Server geometry types in August 2015 (database schema versions 24 and 25). Each annotation footprint (Location) stores a MosaicShape and a VolumeShape, representing the footprint in section-mosaic and registered-volume coordinate spaces respectively; the scalar centroid columns (X, Y, VolumeX, VolumeY) are retained as persisted computed columns derived from those geometries. Spatial indexes (added August 2015) and region/bounds stored procedures support efficient viewport and radius queries during connectome tracing, review, and 3D reconstruction.

How to reproduce these dates

```
# Primary introduction of the spatial columns
git log origin/dev --format="%h %ad %an %s" --date=short \
  -S "MosaicShape geometry" -- "**/CreateUpdateDatabase.sql"

# Inspect the follow-up commits
git show -s --format="%h %ad %an %s" --date=short b579196a 19b70b34 642035
```